EXERCISE 1: UTILIZATION OF A CONSTRAINED RESOURCE

Forefront Tech (FF) is a chip manufacturer, with a maximum factory manufacturing capacity of 20,000 machine hours per week. Factory capacity is a constrained resource.

FF can produce and sell either Chip 123 or Chip XYZ.

	Chip 123	**Chip XYZ**
Prime cost per unit	$20	$15
Processing time (in machine hours) per unit	4	2
Direct labour hours per unit	0.05	0.05
External weekly demand (in units)	2,500	6,000
Selling price per unit	$80	$50

Variable manufacturing overhead cost is $4 per machine hour.

Fixed manufacturing overheads are $100,000 per week allocated based on machine hours.

How many units of Chip 123 and/or Chip XYZ should FF produce and sell to maximum its profits?

EXERCISE 2: UTILIZATION OF A CONSTRAINED RESOURCE

Colonial Heritage makes reproduced antique chairs and tables from select hardwoods.

The following information about its products are as follows:

	Chairs	Table
Selling price per unit	$80	$400
Total variable cost per unit	$30	$200
Hardwood, number of feet per unit	2	10
Monthly demand	600	100

Historically, Colonial's hardwood supplier has been able to supply 3,000 feet of hardwood at $10 per feet.

Due to supply chain issues, Colonial's supplier stated that they will only be able to supply 2,000 feet of hardwood this month.

With the new supply, determine the number of chairs and/or tables Colonial should produce and sell to maximize profits.

EXERCISE 3

CASE: VROOM AUTOS

Vroom Autos manufacture car parts. Car part Alpha ("Alpha") is manufactured by Division A.

1 unit of Alpha requires 1 unit of Subpart Beta ("Beta") that is manufactured by Division B.

	Alpha	**Beta**
Market selling price per unit	$370	$210
Variable product cost per unit	$100 (excl. cost of Beta)	$140
Fixed product cost per unit (allocated based on machine hours)	$30	$20

Division A can purchase Beta externally for $190 per unit, or purchase internally from Division B.

EXERCISE 3A

Market demand and capacity per month are as follow.

In units	**Alpha**	**Beta**
Market demand	5,000	4,000
Manufacturing capacity	3,000	8,000

What will be the range of transfer price Division A and Division B are willing to transact?

EXERCISE 3B

Market demand and capacity per month are as follow.

In units	Alpha	Beta
Market demand	5,000	4,000
Manufacturing capacity	3,000	**4,000**

If purchased externally, Division A will incur quality inspection costs on Beta of $90,000 per month.

What will be the range of transfer price Division A and Division B are willing to transact?

EXERCISE 3C

Market demand and capacity per month are as follow.

In units	Alpha	Beta
Market demand	5,000	4,000
Manufacturing capacity	3,000	**5,000**

Assuming that partial fulfilment is not allowed (i.e. must fulfil all 3,000 units), what is the minimum price Div B will charge Div A (round to 1 decimal point)?